

Projectors

About



Currently two models of projector are used to present stimuli within the FIL imaging facilities.

The projector model and configuration varies depending on the environment in which it is installed

Please study the below [specifications](#) for each installation.

User Guide

For guidance on how to use the projectors see the [Imaging suite User Guide](#).

Projector Specifications

Projector Specifications						
Projector Model	Imaging Facility	Type	Connection	Refresh Rate	Latencies	Native Resolution
Epson EB-L1100U Epson ELPLL08 lens 	3T Quattro MRI 3T Penta MRI 7T MRI OPM	Laser	HDMI HDbaseT	60Hz <i>Windows 10 may display as 59.9Hz</i>	Frames: 2.4 Time/ms: 40	1920×1080
VPiXX ProPiXX 	MEG	LCD	HDMI	60Hz <i>Windows 10 may display as 59.9Hz upto 1440Hz</i>	Frames: 1	1920×1080

Projector Specifications and Visual Angles

Image Dimension and Visual Angle						
Quattro 3T MRI						
Resolution	Throw Distance (mm)	Width (mm)	Height (mm)	Distance Eye-to-Screen (mm)	Horizontal Visual Angle (°)	Vertical Visual Angle (°)
1920 x 1200	3955	392	238	550	39.2	24.4
1920 x 1080	3955	392	221	550	39.2	22.7
1024 x 768	3955	325	246	550	32.9	25.2
Penta 3T MRI						
Resolution	Throw Distance (mm)	Width (mm)	Height (mm)	Distance Eye-to-Screen (mm)	Horizontal Visual Angle (°)	Vertical Visual Angle (°)
1920 x 1200	3690	365	227	550	36.7	23.3
1920 x 1080	3690	365	203	550	36.7	20.9
1024 x 768	3690	304	226	550	30.9	23.2
7T MRI						
Resolution	Throw Distance (mm)	Width (mm)	Height (mm)	Distance Eye-to-Screen (mm)	Horizontal Visual Angle (°)	Vertical Visual Angle (°)
1920 x 1200	3100	355	220	1130	17.9	11.1
1920 x 1080	3100	355	200	1130	17.9	10.1
1024 x 768	3100	314.5	N/A	1130	N/A	N/A
NOTE: Due to restrictions imposed by the head coils used with 7T, the viewable projected image is slightly obscured. See the ' Dummy Screen for 7T '.						
MEG Suite						
Resolution	Throw Distance (mm)	Width (mm)	Height (mm)	Distance Eye-to-Screen (mm)	Horizontal Visual Angle (°)	Vertical Visual Angle (°)
1024 x 768	2500	378	278	550	37.9	28.4
1920 x 1080	2500	482	272	550	47.3	27.8
OPM Suite						
Resolution	Throw Distance (mm)	Width (mm)	Height (mm)	Distance Eye-to-Screen (mm)	Horizontal Visual Angle (°)	Vertical Visual Angle (°)
1920 x 1200	N/A	N/A	N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A	N/A	N/A	N/A

MEG Gamma Correction

Psychtoolbox gamma correction - for use in MEG with ProPixx / VPixx projector **ONLY**

```
%% ProPixx gamma correction example
% Opens a screen on the 2nd monitor (if available)
% corrects gamma
% Closes screen
```

```

%% PsychToolbox basic setup
PsychDefaultSetup(2);                                % apply common Psychtoolbox
parameters                                           % suppress warnings about
Screen('Preference', 'SkipSyncTests', 1);           % suppress warnings about
VBL timing

%% Screen open & setup
scn = max(Screen('Screens'));                        % find second screen if
connected                                           % open a display window
[pWin,wRect] = Screen('OpenWindow',scn);

%% Create & load corrected CLUT
BackupCluts
propixxclut = linspace(0,1,256).^2.2*[1 1 1]; % using a gamma of 2.2
[oldtable1, success] = Screen('LoadNormalizedGammaTable', pWin,
propixxclut);
% ProPixx projector will now behave with the same black, white, & grey
% brightnesses as a standard projector or display

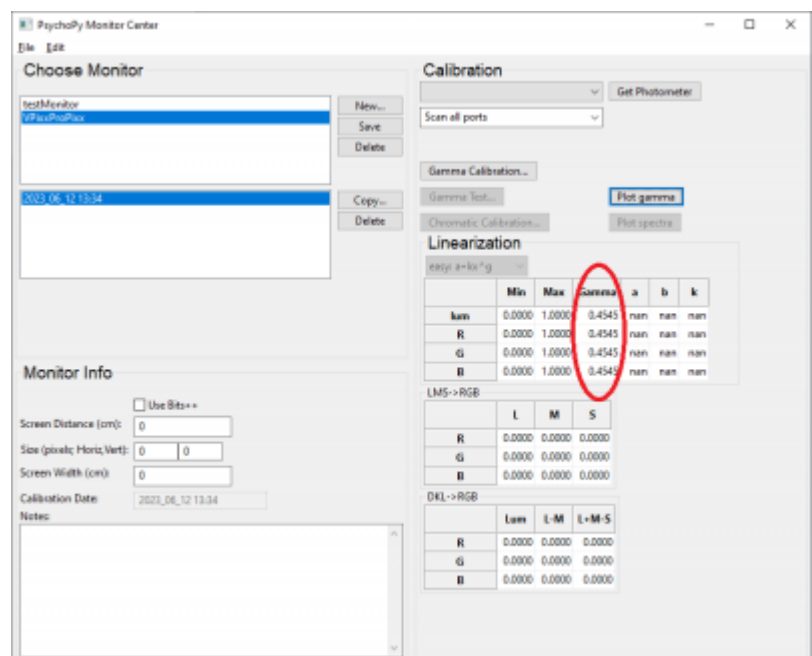
%% Rest of stimuli presented here...

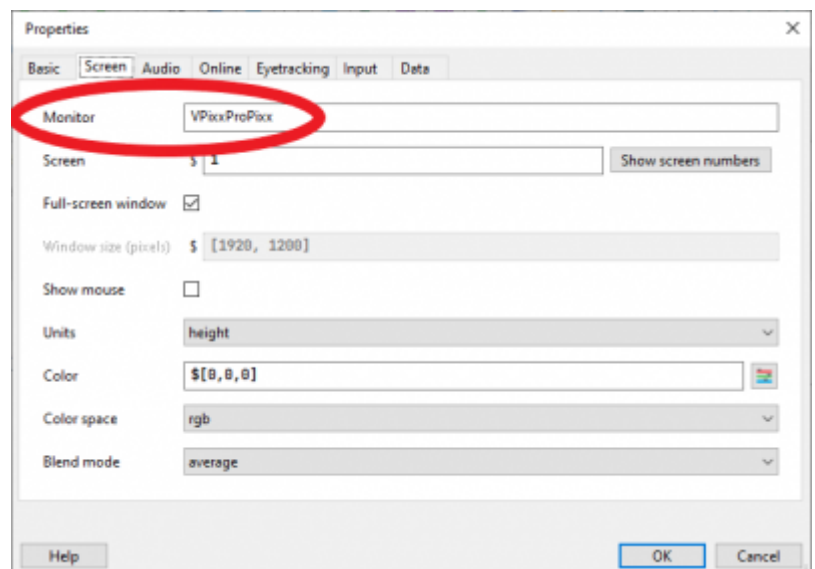
%% Restore original CLUT, tidy up & end
RestoreCluts;
Screen('CloseAll')

```

Psychopy gamma correction - for use in MEG with ProPixx / VPixx projector **ONLY**

Change the gamma values to 0.454545 (i.e. $1/2.2$) as shown and save with a meaningful name. Select the saved monitor setting when you run your experiment.





AND THEN select that saved monitor in the settings / Screen tab

Some background information on gamma curves https://en.wikipedia.org/wiki/Gamma_correction

Troubleshooting

MRI: Unable to Power ON Projector

1. Ensure you are following the correct user guide for your environment.
2. *If using an HDbaseT system:*
 1. Ensure the HDbaseT unit is connected to mains power and switched on.
 2. Ensure the HDbaseT unit is powered on with green LED.
 3. Ensure the remote control is connected to the HDbaseT unit via 3.5mm auxiliary cable.
3. Check mains power to the projector is ON:
 1. Ask an Imaging Support member to switch on mains power.
 1. Penta: Control room switch labelled "Projector Tech RM"
 2. Quattro: Control room switch labelled "Projector MRI 8L1", located via door to the workshop.
 3. MEG:
 4. OPM: N/A

MRI: Blue Screen - No PC Display

The projector is ON, but does not detect an input.

1. Use the remote control to ensure the correct display input is set.
2. Ensure the display Stimulus PC is correctly connected to the projector.
 1. Check display and power connections at:
 1. Stimulus PC
 2. Splitter or HDBaseT unit
3. Ensure the Stimulus PC is outputting correctly to the projector (i.e Extended or duplicate display).

From:

<https://intranet.fil.ion.ucl.ac.uk/wiki/cogentwiki/> - **Cognitive Interface Wiki**

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